Calibration-Free Post-Estimation Fusion of Dual-View 3D Human Pose Streams: Robustness, Front-End Generalization, and Multi-View Extension*

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Abstract

Multi-view capture can resolve the depth ambiguity, self-occlusion, and temporal jitter that limit monocular three-dimensional (3D) human pose estimation, but calibrated multi-view reconstruction is impractical when cameras follow a distant, fast-moving athlete or are installed temporarily. Two views are the smallest deployable multi-view configuration, and modern monocular estimators already deliver an independent 3D pose stream for each view. We present CanonFuse3D, a calibration-free post-estimation framework that fuses two approximately synchronized monocular 3D pose streams directly in pose space. Human-centric canonicalization and sequence-level alignment expose cross-view agreement and disagreement; a shared joint encoder, bidirectional cross-view attention, and temporal refinement then drive joint-frame

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C01 Abstract / 摘要开头

建议从 single-view 直接引出 dual-view

建议调整摘要开头的叙述顺序。当前先从 multi-view 的优势切入，再缩小到 dual-view。虽然两者有包含关系，但这一顺序会使读者先形成广义多视角方法的预期，再转向本文的双视角任务，研究焦点不够集中。建议从 single-view 的局限出发，直接引出增加第二个视角的必要性与互补价值，再介绍 dual-view 场景中的标定和部署困难，最后引出本文的 calibration-free、post-estimation 双视角融合方法。这样主线从 single-view 扩展到 dual-view，与本文核心任务更加一致。

建议的叙述顺序

单视角局限 → 第二视角的互补价值 → 双视角标定与部署困难 → 本文免标定的后估计融合方法

批注为修改建议；原文尚未改写。

reliability-gated fusion with residual correction. The framework is evaluated on three skiing datasets: a synthetic Unity dataset with avatar ground truth, the real-world Ski-PTZ-Pose dataset, and author-collected dual-view recordings for qualitative deployment. Relative to canonical averaging, CanonFuse3D reduces the mean per-joint position error by 44.5% on Unity and 48.6% on Ski-PTZ-Pose and outperforms every non-learned fusion baseline in position, Procrustes-aligned, and acceleration error; a temporal convolutional baseline trained on the same data is competitive in position error, and a temporal smoother attains lower acceleration error at a much higher position error. The gains persist across camera separations, small timing and sampling-rate errors, and structured pose or image occlusion. The same architecture generalizes to alternative pose estimators through lightweight adaptation and extends to additional cameras through pairwise composition.

Keywords: 3D human pose estimation, dual-view fusion, post-estimation pose fusion, calibration-free refinement, human-centric representation, front-end generalization

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1. Introduction

Three-dimensional (3D) human pose estimation from a single camera has advanced rapidly, from temporal lifting of 2D keypoints [1, 2, 3] and pretrained motion representations [4] to video-based body models [5] and foundation-scale full-body estimators that operate on in-the-wild images [6]. These estimators make 3D motion analysis practical for sports, where joint trajectories support performance evaluation, skill assessment, and injury prevention. Their outputs nevertheless inherit the limitations of a single viewpoint: depth along the optical axis is ambiguous, limbs are frequently self-occluded, errors depend on the body orientation relative to the camera, and per-frame predictions jitter over time. Skiing exposes all of these weaknesses at once, because athletes move rapidly, are observed from a distance, and change orientation continuously [7, 8].

Observing the athlete from complementary viewpoints is the established remedy. Calibrated multi-view methods triangulate detections or fuse image features under known camera geometry and achieve high accuracy in laboratories and studios [9, 10, 11, 12, 13]. Field capture of a distant, fast-moving subject breaks their assumptions: cameras are installed temporarily on a slope, pan and zoom to follow the athlete, and cannot be recalibrated before

every session. Weakly calibrated and calibration-free multi-view methods relax the camera requirements [14, 15], but they still reconstruct the pose jointly from images or 2D detections and therefore replace, rather than build on, the monocular estimators that practitioners already run.

These observations motivate a dual-view, post-estimation formulation. Two cameras form the smallest multi-view configuration: they are the cheapest to deploy and synchronize in the field, they already provide a second line of sight against depth ambiguity and self-occlusion, and, as we show later, they capture most of the accuracy available from larger camera groups at a fraction of the fusion cost. Because each view is processed by an off-the-shelf monocular estimator, the available data are two independently estimated 3D pose sequences rather than calibrated image observations. The problem addressed in this article is therefore to fuse two such streams into a more accurate and temporally stable pose sequence without camera intrinsics, extrinsics, or epipolar geometry, at training or inference time.

Figure 1 summarizes the proposed CanonFuse3D framework. Both streams are first expressed in human-centric coordinates defined by body landmarks and aligned to each other by a sequence-level similarity transform, which exposes where the two estimates agree and where they conflict. For every joint and frame, the framework combines the view's own pose, the aligned opposite-view pose, their difference, and motion cues; a shared joint encoder, bidirectional cross-view attention, and short-range temporal refinement then produce a joint-frame reliability gate that weights the two view-specific pose candidates, together with a learned residual correction. The upstream estimator remains separate from the fusion model, so the framework can be placed behind any monocular 3D pose estimator that provides the common joint set.

We evaluate the framework on three skiing datasets: a synthetic Unity dataset with avatar ground truth, the public Ski-PTZ-Pose dataset with a constructed canonical reference, and author-collected indoor and outdoor recordings used for qualitative deployment. Beyond clean-input accuracy, controlled studies examine how the same model behaves when the camera separation, timing, or sampling rate changes, when pose evidence or image regions are missing, when the upstream estimator is replaced, and when more than two cameras are available.

The main contributions are:

- A human-centric framework for calibration-free post-estimation dual-

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C02 1. Introduction / 引言论证主线

从增加第二视角的收益组织引言

引言虽然先介绍单视角局限，后续却转向多视角方案的困难，再缩小到双视角，容易让本文显得是在为多视角系统向下兼容或降低配置。建议以现有单视角估计为起点，围绕增加第二视角能提供哪些互补信息、纠正哪些视角相关误差来建立动机，再解释为什么采用双视角后估计融合，以及如何处理标定和部署约束。这样可以突出增加视角带来的收益，并与摘要后段和后续实验中强调性能提升的论证保持一致。建议重新梳理整章的段落衔接，同时明确每项提升所比较的基线。

建议的叙述顺序

现有单视角估计及其不足 → 增加第二视角的互补收益 → 双视角融合的实际约束 → 本文方法 → 对应的性能提升证据

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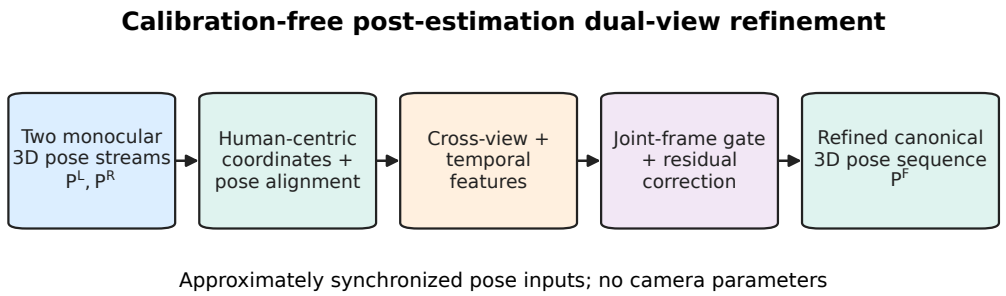


Figure 1: System overview. Two monocular 3D pose streams are expressed in human-centric coordinates, aligned in pose space, and fused using cross-view and temporal features. CanonFuse3D produces a refined canonical pose sequence without camera parameters.

view 3D pose fusion, combining landmark canonicalization, sequence-level alignment, cross-view attention, temporal refinement, and reliability-gated residual fusion.

- Quantitative evaluation on two skiing datasets with explicit reference regimes against non-learned, learned, and calibrated fusion references, on which CanonFuse3D outperforms every non-learned baseline in position, Procrustes-aligned, and acceleration error, complemented by real-world deployment examples.

- A systematic robustness analysis covering camera separation, timing and sampling-rate errors, and both pose-level and image-level occlusion, which identifies the useful operating range and its failure cases.

2. Related Work

2.1. Monocular 3D Human Pose Estimation

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Monocular 3D human pose estimation has progressed from temporal lifting models to transformer-based architectures and large-scale motion representation learning. VideoPose3D uses dilated temporal convolutions to lift 2D keypoint sequences into 3D poses [1]. PoseFormer models spatial joint relations and temporal correlations with transformers [2]. MHFormer represents multiple plausible pose hypotheses before synthesizing a final estimate [3]. MotionBERT pretrains a motion encoder on heterogeneous data

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C03 2.1 / 单目三维姿态相关工作

补足 2D→3D lifting 的分类与梳理

建议加强对 2D→3D lifting 路线的梳理。本节已有 temporal lifting 的概述，也明确介绍了 VideoPose3D 将 2D 关键点序列提升为 3D 姿态，因此问题主要是分类不够清楚、综述不够展开。建议明确区分先检测 2D 关键点再 lifting 的两阶段方法，以及直接从图像或视频估计三维人体的路线；分别概括代表方法、输入和局限，再说明它们如何为本文提供独立的单目 3D 姿态流。避免将不同路线混在同一串模型介绍中。

建议补充重点

2D 检测 → 2D-to-3D lifting；直接从图像或视频估计三维人体；各类方法与本文上游输入的关系。

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for transfer to several human-centric tasks [4]. Recent full-body systems broaden the available front ends: SAM 3D Body estimates body, hand, and foot pose and shape from a single image and emphasizes generalization to diverse in-the-wild conditions [6], while VIBE uses a sequence-level adversarial motion prior to obtain kinematically plausible pose and shape estimates from video [5]. These methods improve the quality of an individual monocular stream, but they do not use a second synchronized stream to correct view-specific errors.

In sports applications, the monocular problem is harder because athletic movements are rapid, atypical of common training corpora, and frequently self-occluded. Closest to our setting, Rhodin et al. introduced the SkiPose PTZ-Camera dataset of alpine skiers observed by rotating cameras and used multi-view consistency as supervision for a monocular model [7], and SportsCap targets monocular sports videos with sub-motion priors [8]. Both remain monocular at inference; complementary 3D predictions from simultaneously available viewpoints are not fused after estimation.

2.2. Multi-view 3D Pose Estimation

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A recent survey of markerless multi-view 3D human pose estimation emphasizes the benefits of complementary viewpoints while identifying occlusion, camera variation, limited supervision, temporal consistency, view selection, and the accuracy–complexity trade-off as continuing concerns [9]. Classical and learned multi-view pipelines usually recover a single 3D pose inside the estimator. Learnable Triangulation combines per-view evidence through differentiable algebraic or volumetric triangulation [10]. Epipolar Transformers transfer image features along epipolar lines in a synchronized, calibrated setting [11], and MVGFormer alternates appearance modules with learning-free geometric modules that triangulate refined 2D poses [12]. AdaFuse addresses unreliable or occluded views by estimating point correspondences between heatmaps and learning camera-specific fusion weights so that poor features do not corrupt stronger views [13]. Although these strategies use different fusion mechanisms, they operate on images, feature maps, heatmaps, or calibrated 2D observations as part of multi-view reconstruction rather than on independently produced 3D pose streams.

Other work relaxes calibration or supervision. MetaPose estimates 3D pose from multiple cameras with unknown calibration, models uncertainty in the input 2D keypoints, and requires only 2D keypoint data for training [14]. FLEX removes the need for extrinsic parameters by predicting view-invariant

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C04 2.2 / 多视角姿态相关工作

补充与双视角任务直接相关的方案

本节对广义多视角方案的梳理可以保留，但本文核心任务是双视角姿态融合，建议补充专门针对双视角或双目设置的代表性方案。比较时需说明相机标定与同步假设、输入是图像或 2D 关键点还是独立 3D 姿态，以及融合发生在哪一阶段。对于可用于双视角的通用多视角方法，也应说明其与本文双视角设置的实际关系，使最接近本文任务的研究得到充分讨论。

建议补充重点

双视角代表方法；标定与同步要求；输入和融合阶段；与本文双视角后估计融合的联系与区别。

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skeletal rotations and bone lengths from multiple video streams and fusing their learned features into one temporally coherent motion [15]. These methods substantially reduce camera requirements, but they still couple cross-view fusion to the multi-view reconstruction network. The setting studied here is complementary: the upstream estimators have already produced two monocular 3D sequences, and the refinement stage receives neither images nor 2D detections, camera parameters, or epipolar geometry.

Takahashi et al. [16] additionally recover temporal synchronization: using detected 2D joints as correspondences, they jointly estimate camera extrinsics, inter-camera time offsets, and 3D joint positions from unsynchronized, uncalibrated cameras. Synchronization is therefore recoverable from human motion itself. In this work the two input streams are assumed to be approximately synchronized, and the remaining timing tolerance is characterized experimentally.

2.3. Pose-Space Refinement, Reliability, and Generalization

Post-estimation refinement provides another way to improve predictions without replacing the upstream estimator. STRIDE is a model-agnostic test-time training method that fits a motion prior to each video and refines noisy initial 3D estimates into temporally coherent poses under occlusion [17]. Its objective is single-video temporal refinement rather than the reconciliation of synchronized viewpoints. Reliability weighting also appears within multi-view estimators: Learnable Triangulation predicts confidence weights for algebraic triangulation, and AdaFuse estimates adaptive weights for camera-view features [10, 13]. In contrast, the reliability gate in this study acts after monocular 3D estimation and selects between aligned 3D pose candidates at the joint–frame level. The distinction is important because upstream confidence, image-feature quality, and residual pose-space reliability are not interchangeable signals.

Robust deployment also depends on how an estimator responds to a changed pose or camera distribution. PoseAug learns differentiable changes to posture, body size, viewpoint, and position to diversify paired 2D–3D training poses and improve cross-dataset generalization [18]. AdaptPose instead generates target-like 3D motions and camera orientations from source poses and uses target-domain 2D poses without requiring target 3D labels or camera information [19]. Both approaches adapt the training distribution of a 3D pose estimator. Our experiments address a different transfer boundary: they hold the pose-space fusion architecture fixed, replace the upstream

monocular estimator, and then compare frozen transfer with target-specific adaptation of the refinement model.

These studies leave a complementary problem: fusing independently estimated 3D pose streams after monocular inference. CanonFuse3D addresses this problem through human-centric pose alignment, cross-view temporal reasoning, and joint-frame gating. The supporting experiments examine the conditions under which this refinement remains useful.

3. Proposed Method

3.1. Overview

CanonFuse3D takes independently estimated pose sequences $P^L, P^R \in \mathbb{R}^{T\times J\times 3}$ from two approximately synchronized views, where T is the number of frames and J the number of joints. It predicts a refined canonical pose sequence P^F of the same shape without camera parameters. Figure 2 shows the complete pipeline. Stages 1–3 place both streams in human-centric coordinates and align them to each other in pose space; stage 4 assembles, for every joint and frame, the five three-dimensional cues that describe the view's own estimate, the aligned opposite estimate, their disagreement, local motion, and the alignment rotation; stages 5–8 encode these cues, exchange information across views and over time, and fuse the two view-specific pose candidates through a reliability gate and a residual correction.

3.2. Human-Centric Landmark Canonicalization

Independent monocular estimates can differ in global position and orientation. We express each stream relative to a fixed reference pose A using the hip midpoint, hip direction, and neck direction. Let h_L, h_R, n denote the left hip, right hip, and neck, and let $\mathrm{nrm}(v) = v/(\|v\|_2+\epsilon)$. The origin and body axes are

$$c = \tfrac{1}{2}(A_{h_L} + A_{h_R}), \quad e_x = \mathrm{nrm}(A_{h_R} - A_{h_L}),$$
$$e_y = \mathrm{nrm}(A_n - c), \quad e_z = \mathrm{nrm}(e_x \times e_y).$$

An additional landmark-direction cue resolves the sign of e_z. With the axis matrix $R = [e_x, e_y, e_z]$, the transformed coordinates are

$$\tilde{P}_{t,j} = (P_{t,j} - c)R.$$

The left stream, right stream, and supervision reference use separate body references. Each origin and axis matrix remains fixed over the sequences

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C05 第 1、2 章 / 问题定义衔接

统一前两章末尾的问题提出与定义

建议对齐第一章的问题提出与第二章末尾的研究空缺表述。第一章明确讨论两路单目 3D 姿态及免标定融合，第二章结尾则概括为融合独立估计的 3D 姿态流，对双视角范围和输入约束的重申不足。两处应围绕同一问题展开：在现有单目估计基础上，利用第二视角的互补信息改善姿态精度与时间稳定性。建议统一写明两路独立估计且大致同步的 3D 序列、后估计融合位置，以及不使用图像、2D 检测和相机参数的融合输入边界，并保持相同的输出与目标表述。

建议统一的问题定义

两路大致同步的单目 3D 序列 → 无图像、2D 检测和相机参数的后估计融合 → 更准确、时间上更稳定的姿态序列。

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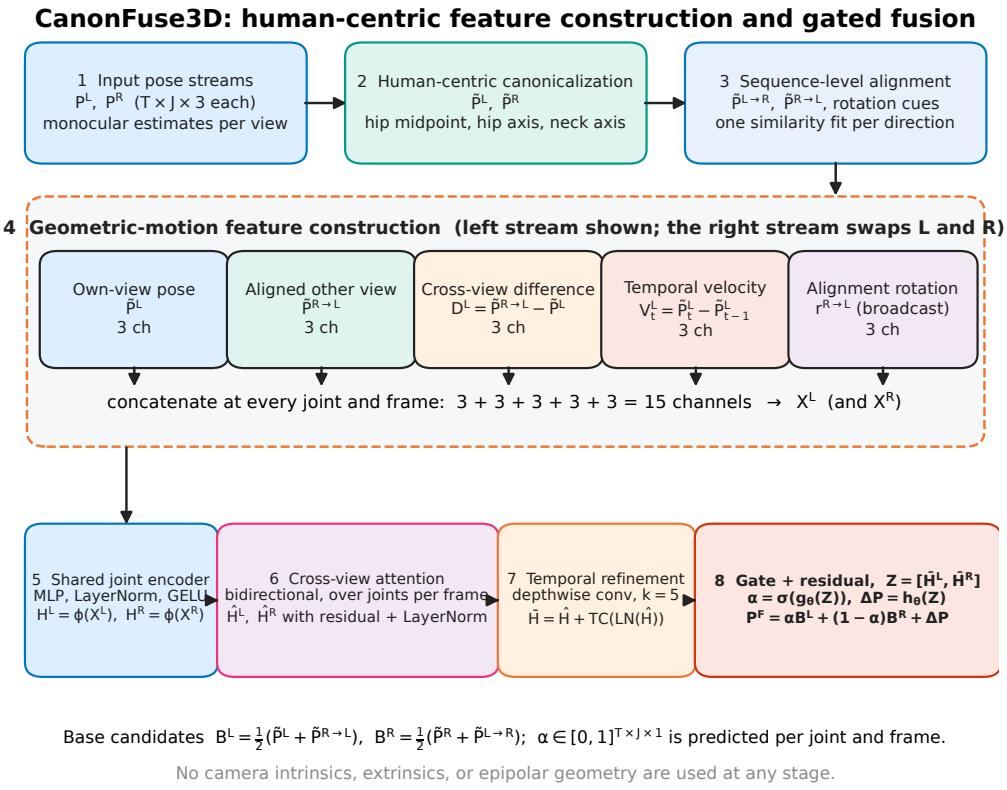


Figure 2: CanonFuse3D pipeline and feature construction. Stages 1–3 canonicalize and align the two pose streams; stage 4 concatenates five three-dimensional cues (own-view pose, aligned opposite-view pose, their difference, temporal velocity, and alignment rotation) into a 15-channel feature at every joint and frame; stages 5–8 encode the features, apply bidirectional cross-view attention and temporal refinement, and fuse the two base candidates B^L and B^R through a joint-frame gate α with a residual correction ΔP. The right stream uses the same construction with the view roles reversed.

to which it is applied; the representation is therefore not a per-frame root-relative pose. Skeleton scale is not normalized, and the constructed axes need not be orthogonal. In the evaluated protocol, the reference is shared within each batch. Its exact selection and landmark conventions are specified in Supplementary Section S9.

3.3. Pose-Level Cross-View Alignment

Because the independently anchored streams can retain residual translation, orientation, and scale mismatch, we apply pose-level cross-view align-

ment in both directions. Specifically, the left pose sequence is aligned to the right pose sequence, and the right pose sequence is aligned to the left pose sequence:

$$\tilde{P}^{L\to R} = \mathcal{A}(\tilde{P}^{L}, \tilde{P}^{R}), \qquad \tilde{P}^{R\to L} = \mathcal{A}(\tilde{P}^{R}, \tilde{P}^{L}),$$

where $\mathcal{A}(\cdot)$ denotes sequence-level pose alignment.

For each sample, let $X, Y \in \mathbb{R}^{N\times 3}$ contain the source and destination joint coordinates in row-vector form, with $N = TJ$. We estimate one uniform scale s, proper rotation Q, and translation d for the whole sequence:

$$(s^\star, Q^\star, d^\star) = \underset{s\geq 0,\ Q\in SO(3),\ d\in\mathbb{R}^3}{\arg\min} \frac{1}{N} \left\| sXQ^\top + \mathbf{1}d^\top - Y \right\|_F^2 .$$

We solve this unweighted least-squares problem by singular value decomposition, excluding reflections. Each direction uses one transform for all frames and joints in the sequence. These auxiliary pose-space transforms do not estimate camera extrinsics; they are also distinct from the per-frame reference alignment used only for evaluation.

The model requires dense, finite pose tensors and receives no joint-validity mask. Missing detections are represented by zero-filled raw poses, and the controlled pose-corruption study replaces or perturbs selected raw entries before canonicalization. All transformed entries participate equally in alignment and training, including corrupted or zero-filled entries; missing joints are neither interpolated nor omitted. Detection-failure flags are used only for reporting, and supervision references remain uncorrupted. Supplementary Section S10 gives the intervention rules.

We further compute cross-view residuals:

$$D^L = \tilde{P}^{R\to L} - \tilde{P}^{L}, \qquad D^R = \tilde{P}^{L\to R} - \tilde{P}^{R}.$$

These residuals encode disagreement between the two pose streams and provide cues about view-specific errors.

3.4. Geometric-Motion Feature Construction

As shown in stage 4 of Figure 2, for each view we construct joint-level geometric-motion features by concatenating pose, cross-view, and temporal cues. For the left-view stream, the feature is defined as

$$X^L = \left[\tilde{P}^{L}, \tilde{P}^{R\to L}, D^L, V^L, r^{R\to L}\right],$$

and for the right-view stream,

$$X^R = \left[\tilde{P}^R, \tilde{P}^{L\to R}, D^R, V^R, r^{L\to R}\right].$$

The velocity terms are first temporal differences of the canonical poses, $V_t^L = \tilde{P}_t^L - \tilde{P}_{t-1}^L$ with $V_1^L = 0$, and likewise for V^R. The terms $r^{R\to L}$ and $r^{L\to R}$ are the rotation vectors of the corresponding sequence-level alignment rotations, broadcast to all joints and frames of the sequence. Each cue is three-dimensional, so every joint at every frame is described by a 15-dimensional feature vector. The five cues give the network the view's own estimate, the opposite view expressed in the same pose space, the residual disagreement between them, local motion, and the global relation between the two streams; together they are the evidence from which the reliability of each candidate is inferred.

3.5. Joint Feature Encoding

The left and right geometric-motion features are passed through a shared joint-wise encoder:

$$H^L = \phi(X^L), \qquad H^R = \phi(X^R),$$

where $\phi(\cdot)$ is a two-layer multilayer perceptron, each layer followed by layer normalization and GELU activation. The encoder maps the 15-dimensional input feature into a 128-dimensional hidden representation for each joint and frame.

Because the encoder is applied at the joint-feature level, it preserves the temporal and skeletal structure of the input sequence while producing a richer representation for subsequent cross-view interaction.

3.6. Bidirectional Cross-View Attention

To exchange complementary information between the two pose streams, we use bidirectional cross-view attention. For each frame, the left-view joint features attend to the right-view joint features, and the right-view joint features attend to the left-view joint features:

$$\hat{H}^L = \mathrm{LN}\Big(H^L + \mathrm{Attn}(Q = H^L, K = H^R, V = H^R)\Big),$$

$$\hat{H}^R = \mathrm{LN}\Big(H^R + \mathrm{Attn}(Q = H^R, K = H^L, V = H^L)\Big).$$

The attention is multi-head and is applied over the J joints within each frame, with separate parameters for the two directions. This allows the network to compare corresponding and related joint features across the two views. When a joint is unreliable in one view, the other view may provide complementary information. The residual connection and layer normalization stabilize the cross-view feature update.

3.7. Temporal Refinement

Frame-wise fusion alone may reduce spatial error but can still produce temporally unstable pose trajectories. To address this issue, we apply a temporal refinement module after cross-view attention.

The temporal refiner operates on each joint's hidden feature trajectory. The sequence is normalized and passed through a depthwise one-dimensional convolution of kernel size 5 along time, a GELU activation, dropout, and a pointwise convolution:

$$\bar{H} = \hat{H} + \mathrm{TempConv}(\mathrm{LN}(\hat{H})).$$

This residual formulation preserves the original cross-view representation while adding local temporal context. The temporal module is lightweight and targets short-range motion consistency, which is important for reducing jitter in sports pose sequences. The same module is applied to $\hat{H}^L$ and $\hat{H}^R$.

3.8. Reliability-Gated Fusion and Residual Correction

After cross-view attention and temporal refinement, the left and right hidden features are concatenated:

$$Z = [\bar{H}^L, \bar{H}^R].$$

A gate head predicts a joint-frame reliability gate:

$$\alpha = \sigma(g_\theta(Z)),$$

where g_θ is a two-layer perceptron with GELU activation, σ is the logistic function, and $\alpha \in [0,1]^{T\times J\times 1}$. The gate controls the relative contribution of the two view-specific pose candidates for each joint and frame.

The two streams have separate body references. The network combines their view-specific candidates through a learned pose-space mapping; this combination does not imply a shared physical coordinate frame.

We construct two base pose candidates:

$$B^L = \frac{1}{2}(\tilde{P}^L + \tilde{P}^{R\to L}), \qquad B^R = \frac{1}{2}(\tilde{P}^R + \tilde{P}^{L\to R}).$$

The reliability-gated base fusion is then computed as

$$P^{base} = \alpha B^L + (1-\alpha)B^R.$$

Finally, a residual head with the same two-layer structure predicts a pose correction:

$$\Delta P = h_\theta(Z),$$

and the final refined pose is

$$P^F = P^{base} + \Delta P.$$

The gate α is interpreted as a learned reliability signal rather than a calibrated probabilistic uncertainty estimate. This distinction is important because the model is trained to improve fusion accuracy and temporal stability, not to produce calibrated uncertainty values.

3.9. Training Objective

The model is trained with a supervised pose loss and four lightweight regularization terms:

$$\mathcal{L} = \mathcal{L}_{sup} + \lambda_{view}\mathcal{L}_{view} + \lambda_{bal}\mathcal{L}_{bal} + \lambda_{ent}\mathcal{L}_{ent} + \lambda_{temp}\mathcal{L}_{temp}.$$

The supervised term is the mean absolute error to the independently canonicalized reference $\tilde{G}$,

$$\mathcal{L}_{sup} = \frac{1}{3TJ}\sum_{t,j} \|P^F_{t,j} - \tilde{G}_{t,j}\|_1.$$

The view-reconstruction term preserves the discrepancy between the two canonicalized inputs. With $\delta = \tilde{P}^L - \tilde{P}^R$, the fused pose and gate reconstruct the two inputs as $\hat{P}^L = P^F + (1-\alpha)\delta$ and $\hat{P}^R = P^F - \alpha\delta$, and

$$\mathcal{L}_{view} = \frac{1}{2}\left(\frac{1}{3TJ}\sum_{t,j}\|\hat{P}^L_{t,j} - \tilde{P}^L_{t,j}\|_1 + \frac{1}{3TJ}\sum_{t,j}\|\hat{P}^R_{t,j} - \tilde{P}^R_{t,j}\|_1\right).$$

This term constrains the gate and fused estimate without treating the sequence-level similarity transforms as physical camera geometry. Two gate penalties avoid degenerate fusion: the balance term $\mathcal{L}_{bal} = |\bar{\alpha} - 0.5|$, where $\bar{\alpha}$ is the mean gate over all joints and frames, discourages collapse to a single view, and the negative-entropy term

$$\mathcal{L}_{ent} = \frac{1}{TJ} \sum_{t,j} [\alpha_{t,j} \log \alpha_{t,j} + (1 - \alpha_{t,j}) \log(1 - \alpha_{t,j})]$$

discourages overconfident gates during training. Finally, the temporal term penalizes second-order variation of the fused trajectory,

$$\mathcal{L}_{temp} = \frac{1}{(T-2)J} \sum_{t=2}^{T-1} \sum_{j} \| P^F_{t+1,j} - 2P^F_{t,j} + P^F_{t-1,j} \|_2,$$

encouraging stable motion without removing the supervised accuracy objective. The weights are $\lambda_{view} = 0.05$, $\lambda_{bal} = 0.02$, $\lambda_{ent} = 0.005$, and $\lambda_{temp} = 0.01$. All sums are additionally averaged over the mini-batch. No term uses an input-visibility mask to omit or reweight missing joints: an available reference still supervises the output at every joint, while a zero-filled input remains part of the view-reconstruction target.

4. Experimental Protocol

4.1. Experimental Setup

Each video view is processed independently by an upstream monocular pose estimator. CanonFuse3D and the pose-only baselines receive the resulting 3D sequences without images, 2D keypoints, or camera parameters. A separate geometric comparison group additionally receives estimated 2D joints and supplied camera calibration. All comparisons use matched sequences, joint conventions, and the stated human-centric coordinate protocol within each dataset.

4.2. Data Sources and Annotation

We use three skiing datasets with complementary roles: two for quantitative evaluation and one for qualitative deployment. Table 1 summarizes their content, scale, and reference regimes; Figure 3 shows paired-view examples.

Table 1: Data sources used in this study, with their content, scale, split, and reference regimes. Split counts are action-disjoint camera-pair sequences for Unity and camera-pair entries for Ski-PTZ-Pose; the author-collected videos are not used for training or quantitative evaluation.

Data source	Content	Scale	Split			Reference / role
			Train	Val	Test	
Unity synthetic	Rendered dual-view videos, monocular 3D pose streams, and the avatar skeleton	2 avatars × 12 actions (24 sequences of 20–65 frames); 180 synchronized views per action (5 layers × 36 azimuths)	193,320	128,880	64,440	Avatar 3D skeleton, 15 joints; training, testing, and all controlled studies
Ski-PTZ-Pose	Real giant-slalom videos with 2D/3D annotations and PTZ calibration metadata	12 sequences; 6 PTZ cameras; about 10K images	150	–	30	Constructed canonical reference from the annotations, 13 joints; real-world evaluation
Author-collected videos	Indoor and outdoor dual-view recordings with estimated pose streams	6 recordings (12 videos); 360–2,609 frames per view	–	–	–	No metric 3D reference; qualitative deployment only

Unity synthetic dataset. We rendered a virtual skiing scene in Unity [20] with one female and one male avatar performing 12 actions each, including steady skiing, braking, falls, and transitions between motion states. Each action is observed from 180 synchronized viewpoints arranged in five layers, each with 36 azimuths at 10-degree intervals. An experimental sample comprises one camera pair; the model always receives two pose streams. The avatar skeleton supplies 3D ground truth independently of the monocular predictions.

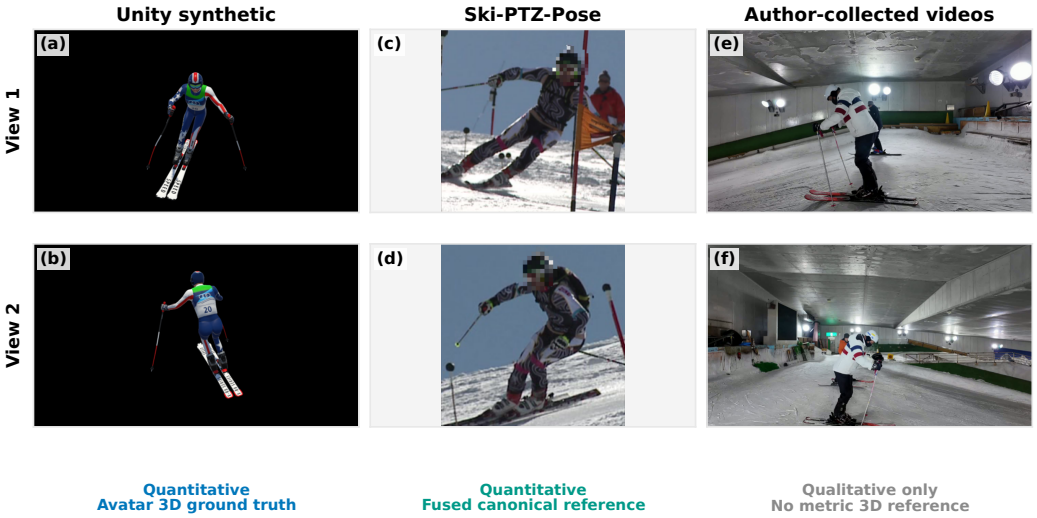


Figure 3: Representative synchronized dual-view examples from the three data sources. Columns show (a,b) Unity synthetic data, (c,d) Ski-PTZ-Pose, and (e,f) author-collected videos; rows show the two input views. Unity and Ski-PTZ-Pose support quantitative evaluation under the reference regimes stated in the text. The author-collected recordings have no independent metric 3D annotations and are used only for qualitative illustration. The Unity panels are foreground-cropped for legibility; the other panels retain their original field of view.

Ski-Pose PTZ-Camera dataset. The public Ski-Pose PTZ-Camera dataset, denoted Ski-PTZ-Pose, contains competitive alpine skiers performing giant slalom runs observed by six synchronized pan-tilt-zoom cameras [7, 21]. It provides 3D poses, projected 2D annotations, and per-frame camera calibration for about 10K images in 12 sequences. For this study, view-specific 3D annotations are aligned to a selected reference camera by similarity transforms fitted on an initial segment; the coordinate-wise median of the aligned annotations on the remaining frames defines the supervision and evaluation reference. Errors therefore measure agreement with this constructed canonical reference, rather than independent physical 3D accuracy. Neither camera calibration nor the annotation-derived transforms enter CanonFuse3D.

Author-collected qualitative videos. Six indoor and outdoor dual-view skiing recordings provide a third dataset for qualitative deployment. A pose estimator processes each view independently. These videos have no independent metric 3D annotations and are excluded from training and quantitative evaluation; they illustrate the visual coherence of the refined motion relative to

Table 2: Joint conventions of the two quantitative datasets. The 15-joint skeleton follows the filtered SAM 3D Body output and the Unity avatar rig. The Ski-PTZ-Pose annotations, like the H36M-style lifting front ends, do not provide the two eye landmarks and are therefore evaluated on the common 13-joint subset.

Joint	Unity avatar	Ski-PTZ-Pose
Left / right eye	✓	–
Left / right shoulder	✓	✓
Left / right elbow	✓	✓
Left / right hip	✓	✓
Left / right knee	✓	✓
Left / right ankle	✓	✓
Right / left hand	✓	✓
Neck	✓	✓
Evaluated joints	all 15	common 13

its input estimates and direct averaging.

Joint convention. The two quantitative datasets are evaluated on different joint sets because their references provide different joints (Table 2). The Unity avatar rig and the native SAM3D front end both provide the 15-joint skeleton, including two eye landmarks that also resolve the facing direction during canonicalization, so Unity is evaluated on all 15 joints. The Ski-PTZ-Pose annotations follow an H36M-style skeleton without eye landmarks, and so do the alternative lifting front ends; the intersection with the 15-joint skeleton is the common 13-joint subset (indices 2–14). Ski-PTZ-Pose and all cross-front-end comparisons therefore use this subset, and the main Unity comparison reports it as well so that both datasets are read on the same joints. The controlled native-input Unity studies use the full skeleton.

4.3. Implementation Details

Each camera-pair sample is uniformly resampled to 30 shared frames. The refinement model uses a shared two-layer joint encoder with hidden dimension 128, four-head bidirectional cross-view attention, a depthwise temporal convolution of kernel size 5, and two-layer gate and residual heads, and is trained with AdamW. Unity uses action-disjoint training, validation, and test splits; Ski-PTZ-Pose uses its predefined training and test partitions. Unity model selection uses validation error, while Ski-PTZ-Pose uses predeclared

checkpoints because no independent validation partition is available. Test errors do not determine model selection. Complete architectures, optimization settings, split sizes, and checkpoint selection details are provided in Supplementary Section S9.

Pose-only comparisons include the two single views, canonical averaging, aligned averaging, a 3D quality-weighted average, a temporal convolutional fusion network (TCN) trained on the same data, and SmoothNet [22] applied to canonical averaging. The separate geometric group includes direct linear triangulation (DLT), reprojection-filtered DLT, and DLT with a learned residual correction. Its additional 2D observations and calibration are stated explicitly in the main comparison table.

4.4. Evaluation Metrics

All errors are reported in dataset coordinate units after the stated body-coordinate transformation, without conversion to millimeters. Lower values indicate better agreement with the dataset's reference. The two reference regimes are analyzed separately.

4.4.1. Mean per-joint position error

Mean per-joint position error (MPJPE) averages Euclidean distances between predicted and reference joint positions:

$$E_{\mathrm{MPJPE}} = \frac{1}{TJ} \sum_{t=1}^{T} \sum_{j=1}^{J} \| P_{t,j}^{F} - \tilde{G}_{t,j} \|_2 .$$

Dataset-level results aggregate all evaluated joint-frame points. No additional reference alignment is applied for this metric.

4.4.2. Procrustes-aligned position error

Procrustes-aligned MPJPE (PA-MPJPE) evaluates pose shape after fitting a least-squares similarity transform from prediction to reference independently in each frame:

$$E_{\mathrm{PA}} = \frac{1}{TJ} \sum_{t=1}^{T} \sum_{j=1}^{J} \left\| s_t P_{t,j}^{F} Q_t^{\top} + d_t - \tilde{G}_{t,j} \right\|_2 ,$$

where, for each frame t,

$$(s_t, Q_t, d_t) = \underset{s \geq 0,\, Q \in SO(3),\, d \in \mathbb{R}^3}{\arg\min} \sum_{j=1}^{J} \left\| s P_{t,j}^{F} Q^{\top} + d - \tilde{G}_{t,j} \right\|_2^2 .$$

The fit allows translation, a nonnegative uniform scale, and a proper rotation, with no reflection. Each reported joint subset is aligned separately. Collapsed predictions are mapped to the reference centroid and retained in the average. This reference fit is used only for PA-MPJPE and does not change model inputs, selection, or other metrics.

4.4.3. Acceleration error

Acceleration error measures agreement between second-order pose differences:

$$E_{\mathrm{acc}} = \frac{1}{(T-2)J} \sum_{t=2}^{T-1} \sum_{j=1}^{J} \|\Delta^2 P_{t,j}^{F} - \Delta^2 \tilde{G}_{t,j}\|_2,$$

where $\Delta^2 P_{t,j} = P_{t+1,j} - 2P_{t,j} + P_{t-1,j}$. It measures temporal agreement at the evaluated sampling interval, in coordinate units per sampled frame squared.

4.5. Robustness and Generalization Protocols

All controlled studies use the frozen Unity checkpoint and the complete held-out split; the model is updated only in the explicitly identified adaptation experiments. Camera separation is analyzed in six 30-degree bins of circular azimuth difference between the two cameras. Timing robustness uses seven right-stream offsets from -2 to $+2$ frames, including half-frame conditions, and seven sampling-rate errors from -2% to $+2\%$; an auxiliary diagnostic additionally estimates larger offsets of up to five frames. Pose-level corruption crosses three affected-view modes, three patterns, and five ratios, giving 45 conditions, and image-level occlusion applies the same view modes and patterns at two ratios to the RGB frames before the native pose estimator, giving 18 conditions; the two ratios parameterize different interventions and do not imply equal physical severity. Front-end generalization replaces SAM3D by three lifting estimators applied to the same ground-truth 2D tracks, isolating the lifting stage, followed by frozen transfer and target-specific or mixed adaptation. Finally, the multi-view extension composes the frozen dual-view model over all camera pairs of nested one- to four-camera groups. Full intervention and aggregation rules are given in Supplementary Section S10.

5. Results

5.1. Primary performance and cross-dataset evidence

Table 3 compares CanonFuse3D with the pose-only fusion baselines and with the separately labeled geometric group on the common 13-joint subset of both datasets. On all 64,440 held-out Unity camera-pair sequences, CanonFuse3D reduces MPJPE from 0.2792 for canonical averaging to 0.1551 (44.5%). Canonical averaging is penalized by the weaker right stream (0.4617): aligning the right stream to the left before averaging removes most of this discrepancy (0.1622), and the quality-weighted average brings no further gain. CanonFuse3D improves on the best of these non-learned methods in MPJPE (0.1551 versus 0.1622), PA-MPJPE (0.0637 versus 0.0667), and acceleration error (0.0297 versus 0.0308), so the learned fusion adds accuracy and temporal stability beyond what pose-space alignment alone provides. It is also more accurate than the temporal convolutional baseline trained on the same data (0.1610 MPJPE, 0.0690 PA-MPJPE, 0.0339 acceleration error). SmoothNet applied to canonical averaging gives the lowest acceleration error in the group (0.0113) but leaves the position error at 0.2925, so temporal smoothness and positional accuracy must be read separately. On all 15 Unity joints, the corresponding reduction over canonical averaging is 46.3% (0.1541 versus 0.2868; Supplementary Table S1).

On the 30 fixed Ski-PTZ-Pose test sequences, the two views disagree far more (0.3552 and 0.4462), and pose-space alignment alone does not resolve the disagreement (0.3537). CanonFuse3D reduces MPJPE to 0.1794, a 48.6% reduction over canonical averaging, with PA-MPJPE 0.1269 and acceleration error 0.0311. On this small real-world index the temporal convolutional baseline reaches lower MPJPE and PA-MPJPE (0.1581 and 0.1144) at a higher acceleration error (0.0411), and SmoothNet again gives the lowest acceleration error (0.0151) with a position error close to averaging (0.3227). These values measure agreement with the constructed annotation reference rather than independent physical 3D accuracy, and the 30-sequence index provides limited resolution between the two learned models.

The geometric group quantifies what calibration buys. With the estimated 2D joints and the supplied calibration, plain DLT is less accurate than CanonFuse3D on both datasets (0.1983 and 0.3984), reprojection gating does not improve it, and only a learned residual correction on top of DLT reaches lower MPJPE (0.0964 and 0.1643), at the cost of requiring 2D observations and camera calibration that the pose-only setting excludes. This

Table 3: Main comparison on the common 13-joint subset with the native SAM3D front end. Unity uses all 64,440 held-out camera-pair sequences; Ski-PTZ-Pose uses all 30 fixed test sequences. Lower is better. Bold marks the lowest value within each input group and column.

	Unity			Ski-PTZ-Pose		
Method	MPJPE ↓	PA-MPJPE ↓	Accel. ↓	MPJPE ↓	PA-MPJPE ↓	Accel. ↓
3D pose inputs only						
Left canonical	0.1624	0.0711	0.0355	0.3552	0.1864	0.0531
Right canonical	0.4617	0.0759	0.0452	0.4462	0.1754	0.0455
Canonical average	0.2792	0.1162	0.0322	0.3487	0.1877	0.0392
Aligned average	0.1622	0.0667	0.0308	0.3537	0.1718	0.0381
3D quality-weighted fusion	0.2767	0.1152	0.0312	0.3499	0.1875	0.0385
Temporal fusion (TCN)	0.1610	0.0690	0.0339	**0.1581**	**0.1144**	0.0411
Avg. + SmoothNet [22]	0.2925	0.1860	**0.0113**	0.3227	0.1924	**0.0151**
CanonFuse3D (ours)	**0.1551**	**0.0637**	0.0297	0.1794	0.1269	0.0311
Additional estimated 2D joints and camera calibration						
Calibrated DLT	0.1983	0.1146	0.0376	0.3984	0.2219	0.0816
Reprojection-gated DLT	0.2088	0.1196	**0.0295**	0.3996	0.2234	0.0846
DLT-residual MLP	**0.0964**	**0.0535**	0.0403	**0.1643**	**0.1172**	**0.0598**

group is therefore not a common-input comparison with CanonFuse3D. On Ski-PTZ-Pose the geometric predictions are first centered at their own per-frame pelvis, consistent with the root-relative annotations; this preprocessing fits nothing to the test reference.

5.2. Reliability-gate behavior

C06

The complete-split gate-error correlation is 0.650, and the gate assigns the larger weight to the lower-error view in 91.7% of common-13 joint-frame comparisons. To examine whether the learned gate collapses to a single view or preserves local fusion flexibility, Table 4 summarizes the predicted fusion weights on the complete held-out Unity split without input masking. The reported value is the mean left-candidate weight α, averaged over frames and joints within each body region; $1 - \alpha$ is the corresponding right-candidate weight. The learned gates remain close to balanced on average, which is consistent with the balance regularization in the training objective, while retaining small region-dependent variations. Together with the gate–error correlation and view-preference accuracy, this indicates that the gate carries local reliability information without constituting calibrated uncertainty.

审阅批注

原文第 20 页 · 用户意见整理

C06 5.2 / 可靠性门控验证

补充极端条件下的门控响应与消融

现有相关性和偏好准确率说明权重与输入误差有关，但尚不足以单独证明动态门控对最终性能的贡献。平均权重接近 0.5 也不能判断逐关节逐帧的权重是否接近均匀。建议在一侧严重受损、另一侧可靠，并交换受损侧或使可靠侧随时间切换的条件下，报告门控分布、时间变化及受损关节的权重响应。同时与固定 0.5 权重、固定单侧权重和去除动态门控的模型比较，控制其余网络与训练条件，分别说明残差修正和门控的贡献。将这些分析与已有姿态损坏和图像遮挡实验联系起来，直接验证门控在需要选择信息时是否有效。

建议验证重点

单侧极端受损及左右互换；可靠侧随时间切换；逐关节逐帧权重分布；固定权重与动态门控的受控比较。

批注为修改建议；原文尚未改写。

Table 4: Region-wise reliability-gate statistics on the complete Unity test split without masking.

Region	Joints	α	$1-\alpha$
Upper limbs	6	0.497 ± 0.004	0.503
Lower limbs	6	0.502 ± 0.014	0.498
All joints	15	0.497 ± 0.011	0.503

Note. Values report the mean and across-joint standard deviation of per-joint mean α. Balance and entropy regularization act on the gate, so these statistics must not be interpreted as probabilistic calibration.

5.3. Camera-pair separation

The controlled studies in this and the following three subsections vary one capture condition at a time while keeping the checkpoint, held-out split, and evaluation path fixed. The camera-angle breakdown in Table 5 does not reveal a failure regime. CanonFuse3D MPJPE decreases from 0.1563 in the 0–30 degree bin to 0.1528 around 120–150 degrees, while the gain over canonical averaging stays between 45.9% and 46.6%. Wider azimuth separation therefore supplies slightly more complementary evidence in this synthetic setup, but the narrow range cautions against claiming a strong optimum. The bins contain 8,640–12,600 mutually exclusive test camera-pair sequences and together cover the complete 64,440-sequence test split.

To account for the four action-specific observations associated with each camera geometry, we collapse results by unordered camera pair before inference. Every angle bin retains a positive fusion gain under paired Wilcoxon testing (all Holm-adjusted $p < 0.001$; rank-biserial correlation 0.994–1.000). The across-bin Kruskal–Wallis test is statistically significant ($p = 0.025$) but has a negligible effect size ($\epsilon^2 = 0.00049$). Of 15 Holm-adjusted post-hoc comparisons, only 90–120 versus 150–180 degrees remains significant ($p = 0.0064$), again with a small rank-biserial effect ($r_{rb} = 0.053$). Thus, the fusion benefit is reliable within every tested bin, whereas camera separation explains little of its magnitude. Detailed paired-test statistics are provided in Supplementary Table S2; Supplementary Fig. S1 shows the corresponding absolute-error curves.

Table 5: Accuracy and camera-pair bootstrap intervals by azimuth separation. Positive gain favors CanonFuse3D.

Angle (deg)	Pairs	Canon. avg.	Fused	Mean gain	95% CI
0–30	2,160	0.2891	0.1563	0.1328	[0.1313, 0.1343]
30–60	2,700	0.2888	0.1554	0.1334	[0.1321, 0.1347]
60–90	2,700	0.2876	0.1544	0.1332	[0.1318, 0.1345]
90–120	2,700	0.2865	0.1529	0.1337	[0.1324, 0.1349]
120–150	2,700	0.2852	0.1528	0.1325	[0.1314, 0.1336]
150–180	3,150	0.2844	0.1532	0.1312	[0.1303, 0.1322]

Note. Each camera pair contributes four action-specific sequences; the bins cover all 64,440 test sequences. MPJPE and absolute gain use dataset coordinate units. Intervals resample camera pairs. Detailed paired tests and effect sizes appear in Supplementary Table S2.

5.4. Temporal offset and sampling-rate drift

Figure 4 summarizes two synchronization stress tests as relative MPJPE changes from each method's unperturbed value. Supplementary Table S3 provides the absolute errors and gate diagnostics. At zero perturbation, all-15 MPJPE is 0.1541. Across ± 2 frame offsets, it ranges from 0.1537 to 0.1555; the worst case, a two-frame positive offset, is only 0.91% above zero. Gate-error correlation stays between 0.646 and 0.650 and view-preference accuracy remains above 91.4%. The modest variation suggests that sequence-level alignment and the five-frame temporal refiner tolerate short local disagreement, but this experiment does not establish robustness to long delays or unsynchronized action phases.

Clock-rate perturbations are even smaller. At $+2\%$, MPJPE increases by 0.23%; negative perturbations up to -2% change MPJPE by less than 0.02%. Acceleration error is not monotonic because temporal resampling smooths as well as misaligns trajectories. Consequently, MPJPE is the cleaner measure of rate-drift sensitivity in this controlled test.

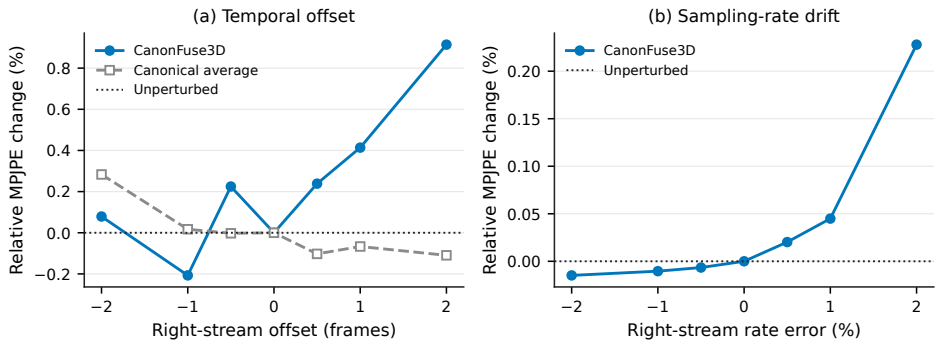


Figure 4: Relative MPJPE change under right-stream temporal offset and sampling-rate error on the complete held-out split. Each curve is normalized by its own method-specific unperturbed MPJPE; positive values indicate increased error. Dotted horizontal lines mark zero change.

5.5. Missing-pose evidence

Figure 5 reports all 45 conditions of the complete held-out-split missing-pose study; Supplementary Table S4 gives exact errors and gains at the representative ratios 0.5 and 1.0. Corrupted entries use a fixed mixture of zero replacement (35%) and Gaussian perturbation (65%, standard deviation 0.08 in raw input coordinate units, before canonicalization). For random masking, the ratio is the independent joint-frame probability; in the both-view condition, masks and corruption noise are sampled separately for the two streams. Distal-subset masking applies that probability only to the same eight filtered joint indices used by the executable masking protocol. Temporal masking selects the stated fraction of joints and corrupts one independently positioned contiguous 10-frame interval for each selected joint within every 30-frame sequence; its ratio therefore denotes selected joints rather than the global fraction of tensor entries.

Single-view corruption tests whether the complementary stream can rescue degraded evidence, whereas both-view corruption probes the method when redundancy is removed. The curves compare fused and canonical-average MPJPE, making explicit where learned fusion retains an advantage and where both methods fail together. At ratio 0.5, single-view temporal masking retains gains of 37.0% for left corruption and 37.9% for right corruption; distal masking retains 28.0% and 31.5%, respectively. The advantage narrows for unstructured random corruption. Most importantly, when random corruption affects both streams at ratio 0.5, fusion is 3.3% worse than

C07

审阅批注

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C07 5.5–5.6 / 信息受损与几何辅助

比较额外几何信息能否改善受损输入

两路输入同时严重损坏时出现退化是可以预期的；损坏比例为 0.5 也不等于完全没有剩余信息。建议进一步研究在相同受损条件下，额外的相机内参、外参及可用二维观测或几何约束能否改善结果。可比较仅姿态融合与几何辅助方案，并报告收益随损坏程度和几何信息质量的变化。表 3 已有带标定的几何比较组，但尚不能回答相同损坏条件下的恢复能力。额外信息应明确列为不同输入设置；相机参数本身并不提供缺失的关节观测，能否改善需要实验验证。

建议扩展实验

匹配样本与受损条件；明确额外可用信息；比较姿态融合与几何辅助；分析恢复收益及失效范围。

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canonical averaging, whereas bilateral temporal and distal corruption still retain gains of 26.9% and 12.5%. Thus, the checkpoint can exploit a clean complementary stream and structured missingness, but dense independent corruption in both inputs lies outside its reliable regime.

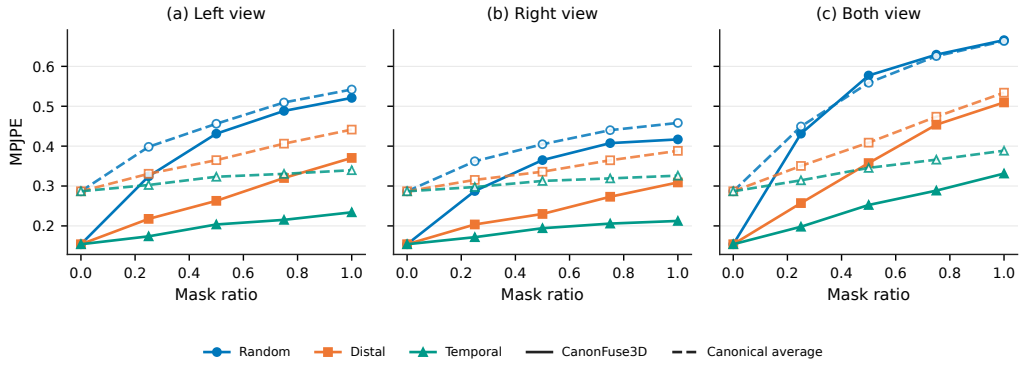


Figure 5: Complete held-out-split masking robustness for left-, right-, and both-view corruption.

5.6. Image-level occlusion through the pose front end

Figure 6 illustrates the three image-level occlusion patterns. Square patches filled with the image mean color are placed on the projected joints selected by the same random, distal, and temporal rules as the pose-level study, with a side length of 12% of the projected person height; the masked frames are then processed by SAM3D and the frozen fusion model, so the resulting errors include missed detections and distorted poses produced by the front end. Supplementary Table S5 lists all 18 conditions.

CanonFuse3D improves on canonical averaging in 15 of the 18 conditions. At random ratio 0.5, gains are 31.9%, 34.3%, and 20.9% for left-, right-, and both-view occlusion, respectively. All twelve distal and temporal conditions retain positive gains, ranging from 14.6% to 43.4% for distal masks and from 42.3% to 45.9% for temporal masks. These results extend the controlled robustness evidence to errors produced by the image-to-pose front end.

Random occlusion at ratio 1.0 reverses the advantage in all three affected-view modes: fused MPJPE exceeds canonical averaging by 18.0%, 9.2%, and 28.4% for left, right, and both views. In the bilateral case, MPJPE is 0.6311 versus 0.4914, and the pair-aligned SAM3D detection-failure rate is 36.36%. Failure rates for the corresponding left- and right-view conditions are

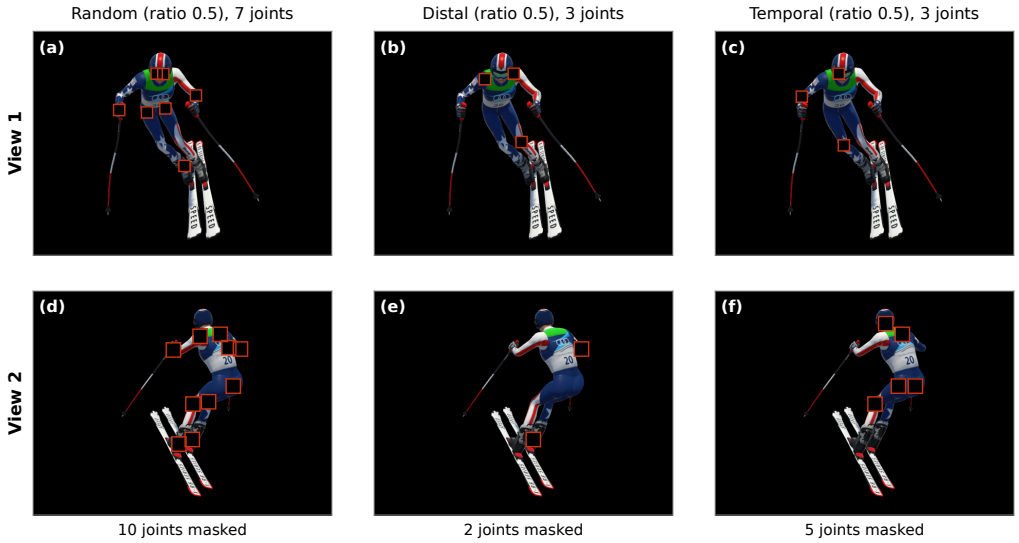


Figure 6: Image-level occlusion examples on one Unity frame. Rows show the two input views and columns the three patterns at ratio 0.5: random joint-frame selection, distal-subset selection, and temporal selection of contiguous 10-frame intervals. Red outlines mark the mean-color patches placed on the selected projected joints; the number of masked joints in each panel is stated in its label. The outlines are drawn for illustration only and are not present in the images passed to SAM3D.

40.51% and 32.22%. Failed detections remain in the evaluation as zero poses, so these results include complete upstream detection loss. The aggregate measurements do not isolate how much of the degradation comes from failed detections, errors in detected poses, or the fusion response. An unoccluded complementary view therefore does not guarantee a fusion advantage under this severe intervention.

Figure 7 compares fused MPJPE for image-level and direct pose-stream interventions. Their matching ratios control different perturbations. For example, bilateral random corruption at ratio 0.5 retains a 20.9% gain in the image-level experiment but gives a 3.3% disadvantage in the pose-level experiment (Supplementary Tables S5 and S4). This difference shows that the two protocols probe distinct error distributions; it does not establish an ordering of physical occlusion severity. The condition counts describe one frozen checkpoint, rather than independent training repetitions.

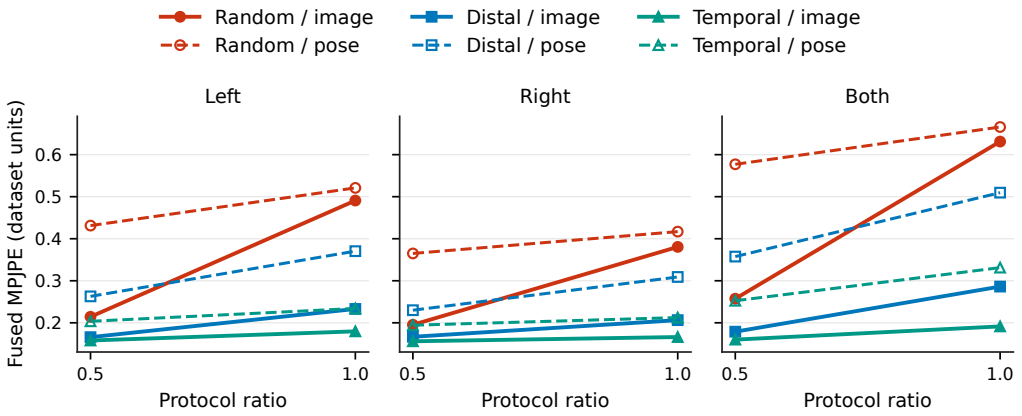


Figure 7: Fused MPJPE under image-level occlusion propagated through SAM3D and direct pose-stream corruption, for left-, right-, and both-view interventions. Solid lines with filled markers denote image-level masks; dashed lines with open markers denote pose-stream corruption. Matching protocol ratios do not imply equal physical occlusion severity. Errors use dataset coordinate units.

5.7. Front-end generalization and adaptation

CanonFuse3D is trained on SAM3D streams, but the framework itself is front-end agnostic: it consumes only 3D pose sequences on the common joint set and can be placed behind any monocular estimator that provides them. Table 6 first evaluates the frozen SAM3D-trained checkpoint on three alternative lifting front ends without any retraining. MotionBERT retains a positive gain of 8.0% over its own canonical average, whereas PoseFormer and VideoPose3D degrade by 18.0% and 12.8%. Canonicalization removes global position and orientation differences but not estimator-specific limb bias and temporal statistics, so a frozen model remains coupled to the error distribution it was trained on, and transfer to a new estimator should be validated rather than assumed.

Lightweight adaptation restores the benefit on every front end. Table 7 reports the complete 8×4 matrix in which the frozen reference checkpoint and seven adapted models are each evaluated on all four front ends; Supplementary Fig. S2 visualizes the corresponding relative recovery. Each adaptation updates either the gate and residual heads only or all parameters for 20 epochs on training and validation actions, without test data. Full target-specific adaptation reduces the matched-front-end error to 0.2290 for VideoPose3D, 0.2741 for PoseFormer, and 0.2393 for MotionBERT, which

Table 6: Frozen transfer to alternative upstream estimators on the common 13-joint subset. Lower MPJPE is better. Gain is relative to canonical averaging; positive values indicate improvement.

Front end	Canon. avg.	Fused	Gain (%)	Gate pref. (%)
MotionBERT	0.4766	0.4385	+8.0	61.4
PoseFormer	0.4542	0.5360	-18.0	52.3
VideoPose3D	0.4776	0.5389	-12.8	47.0

Table 7: Common-13 MPJPE after front-end-specific or mixed adaptation. Each row is one frozen model evaluated on all four front ends; lower is better.

Adaptation model	SAM3D	VideoPose3D	PoseFormer	MotionBERT
Frozen reference	0.1551	0.5389	0.5360	0.4385
VideoPose3D/heads	0.2120	0.2621	0.3297	0.3567
VideoPose3D/full	0.2442	0.2290	0.2954	0.4403
PoseFormer/heads	0.1822	0.2833	0.3234	0.3521
PoseFormer/full	0.2033	0.2447	0.2741	0.3381
MotionBERT/heads	0.1976	0.4667	0.5100	0.2744
MotionBERT/full	0.2117	0.5050	0.6036	0.2393
Mixed/full	0.1809	0.2363	0.2751	0.2464

is 52.0%, 39.7%, and 49.8% below the canonical average of the respective front end (0.4776, 0.4542, and 0.4766) and 57.5%, 48.9%, and 45.4% below the frozen checkpoint on the same input. A single model adapted on a balanced mixture of all four front ends reaches 0.2363, 0.2751, and 0.2464 on the three alternatives while retaining 0.1809 on SAM3D, which is 16.7% above the native checkpoint but still 35.2% below canonical averaging. Thus one architecture, adapted with a short schedule, improves over canonical averaging on every tested front end; the remaining trade-off is between recovery on the target front end and retention on the native one, since no adapted model dominates the frozen checkpoint on every input source.

5.8. Scaling beyond two views

Figure 8 shows accuracy and bootstrap intervals under frozen pairwise composition, while Table 8 reports its computational cost. Supplementary Table S6 provides the numerical means, standard deviations, and intervals for all methods. Accepted and rejected camera groups are retained in the run-

Table 8: Accuracy–cost trade-off for deployable frozen pairwise composition. Latency is the serial fusion time per 30-frame camera group; reduction is relative to the two-view MPJPE.

Views	Pairs	MPJPE	Red. (%)	Latency (ms)	Rel.	Groups/s	Peak MiB
2	1	0.1295	–	2.02	1.00×	494.7	12.28
3	3	0.1177	9.1	6.02	2.98×	166.1	12.31
4	6	0.1158	10.6	12.06	5.96×	83.0	12.33

Serial latency is the synchronized mean over 44 accepted 30-frame camera groups after 10 warm-up executions on one NVIDIA RTX A6000. It includes pose canonicalization, all unique pair forwards, and output averaging, but excludes monocular front-end inference, data loading, and oracle selection.

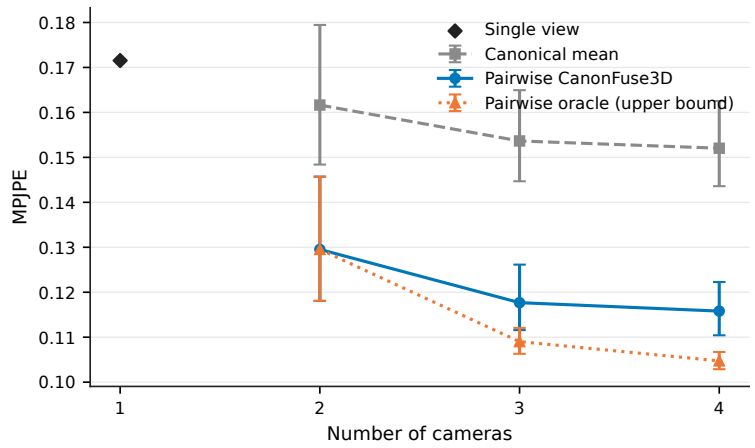


Figure 8: N-view scaling under frozen pairwise CanonFuse3D composition. Error bars show paired-group bootstrap 95% confidence intervals. The oracle is an upper bound.

time manifest so that missing common frames are not silently converted into favorable subsets. The two-view configuration captures the largest gain: it reduces MPJPE from the single-view value of 0.1715 to 0.1295 (24.5%) with one pair forward and 2.02 ms serial fusion latency. Three views reduce MPJPE by a further 9.1% relative to two views, but require three pair forwards and 6.02 ms (2.98-fold) latency. Four views reach 0.1158 MPJPE, a 10.6% reduction relative to two views, while requiring six pair forwards and 12.06 ms (5.96-fold) latency; the incremental gain from three to four views is only 1.6%. Thus, additional views remain useful when accuracy is prioritized, whereas two views provide the most favorable accuracy–cost operating point for the main CanonFuse3D configuration.

C08

审阅批注

原文第 28 页 · 用户意见整理

C08 5.8 / 效率与计算成本

补充模型规模与计算量等效率指标

效率证据仍需补全。表 8 已报告串行融合耗时、吞吐量和峰值显存，但当前 PDF 未明确给出参数总量、可训练参数量、模型存储大小以及 FLOPs 或 MACs。建议按明确的帧数、关节数和视角数报告这些指标，并区分复用同一模型的参数规模与随相机对数量增长的计算量。与主要学习式基线在相同硬件、批大小和数值精度下比较，说明预热、同步及计时范围。融合模块耗时应与包含上游姿态估计的端到端耗时分开报告，以支持轻量化、效率和部署成本的论述。

建议补充指标

参数量与模型大小；指定输入下的 FLOPs/MACs；同条件基线比较；融合与端到端耗时；多视角扩展的计算增长。

批注为修改建议；原文尚未改写。

5.9. Automatic synchronization diagnostic

Figure 9 compares uncorrected, automatically corrected, and oracle streams across the eleven injected offsets. Supplementary Table S7 reports the exact errors together with lag-estimation accuracy and correction-activation rates. The confidence threshold (0.09296) was fixed from validation data before these test results were read. The result is directionally asymmetric. Automatic correction improves MPJPE for every tested positive lag: at +5 frames, it reduces error from 0.1537 to 0.1506, versus the 0.1488 oracle. It also gives smaller improvements at ±0.5 frames. In contrast, it worsens every tested negative integer lead; at −5 frames, error increases from 0.1484 to 0.1498. At zero offset, 41.5% of samples activate a nonzero correction and MPJPE increases slightly from 0.14879 to 0.14884. Exact lag accuracy is only 13.2% at +5 frames. Thus, the descriptor is a useful diagnostic and partially recovers positive delays, but it is not a reliable bidirectional synchronization solution. We do not use it to strengthen the main CanonFuse3D comparison.

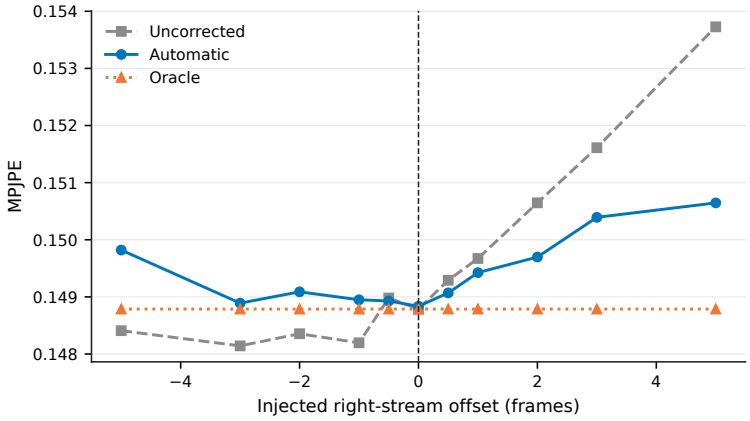


Figure 9: MPJPE under injected right-stream offsets on the fixed central 20-frame interval. Oracle restores the original unperturbed right stream; automatic correction uses the validation-calibrated confidence threshold.

5.10. Qualitative evaluation

Figure 10 illustrates deployment on the author-collected skiing videos, where left- and right-view poses are estimated independently and no camera calibration is used during inference. The proposed result remains more coherent than either single-view prediction or direct averaging, especially

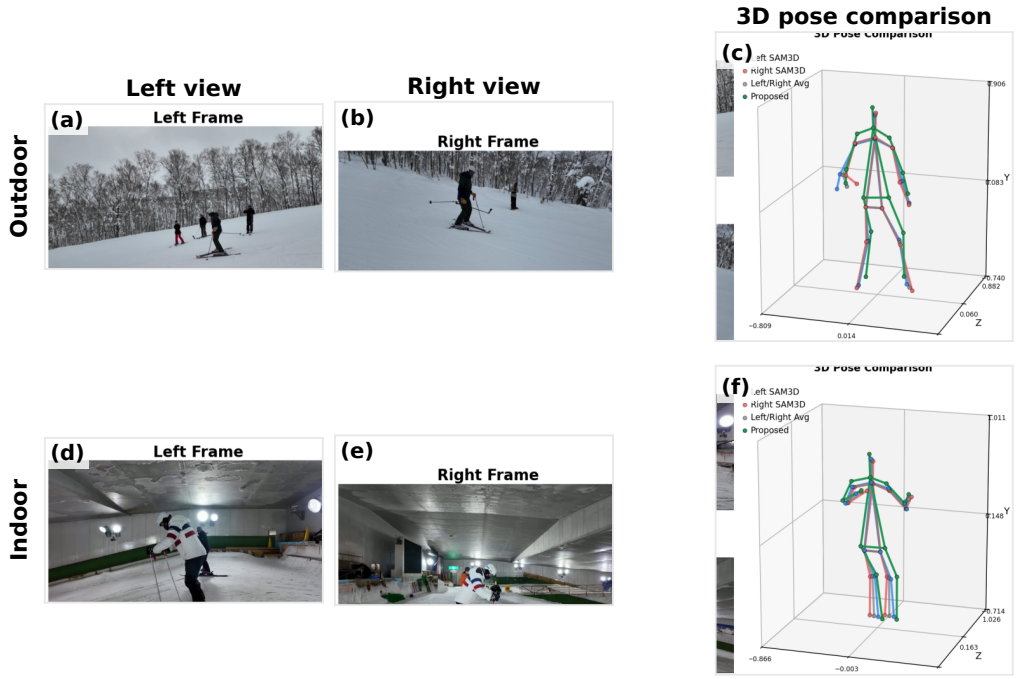


Figure 10: Qualitative outputs on author-collected outdoor and indoor skiing videos. Panels (a), (b), (d), and (e) show the synchronized input views; panels (c) and (f) compare the two single-view predictions, canonical average, and CanonFuse3D output. These videos have no independent metric 3D annotations and are not used for training or quantitative evaluation. These examples were also included in the earlier conference manuscript.

when the two views disagree around arms, knees, and feet. Because no independent 3D ground truth is available, this comparison is qualitative and is not included in any reported accuracy metric. The examples also show the main limitation: if both upstream predictions contain similar errors, the refinement can still inherit part of that error.

Qualitatively, the refined skeleton tends to preserve the more stable limb configuration while avoiding abrupt compromises caused by direct averaging. This is particularly visible in poses with strong torso rotation, occluded lower limbs, or large differences between left- and right-view predictions. For sports analysis, such behavior is useful because downstream measures such as joint-angle trajectories, limb extension timing, and motion smoothness are sensitive to frame-level outliers. At the same time, the examples should be read as a post-estimation refinement result rather than as full 3D reconstruction from images; the method improves the consistency of available pose

streams but does not recover information that is absent from both inputs.

6. Discussion

The main comparison places CanonFuse3D between two groups of references. Among methods that receive only the two 3D pose streams, it outperforms every non-learned baseline on both datasets in position, Procrustes-aligned, and acceleration error, and the margin over pose-space alignment alone shows that the learned gate and residual contribute beyond coordinate normalization. Against learned references the picture is metric-dependent: CanonFuse3D is more accurate than the temporal convolutional baseline on Unity, whereas that baseline reaches lower position error on the 30-sequence Ski-PTZ-Pose index, and SmoothNet attains the lowest acceleration error on both datasets while leaving the position error near canonical averaging. The calibrated group shows what additional information buys: plain triangulation of the estimated 2D joints is less accurate than the pose-only fusion, and only a learned correction on top of it surpasses CanonFuse3D. For field capture in which calibration is unavailable, the pose-only result is the relevant operating point. The perturbation, transfer, and view-scaling studies compare against canonical averaging, the non-learned reference that shares the model's inputs; they characterize the operating range of CanonFuse3D rather than rank fusion architectures under perturbation.

The combined evidence supports a narrower claim than generic calibration-free multi-view reconstruction. CanonFuse3D is effective as a pose-space refinement layer when its two inputs are approximately synchronized, share the training front end's error distribution, and contain complementary evidence. It requires no camera parameters and retains a large advantage across the tested Unity angle bins. Pairwise composition extends the frozen dual-view model to larger camera groups, but it does not learn a joint representation over more than two views. Under the evaluated serial implementation, the number of forwards grows from one pair at two views to three and six pairs at three and four views, increasing latency from 2.02 ms to 6.02 and 12.06 ms. Because two views capture the largest accuracy gain over a single view while additional views provide smaller improvements at approximately threefold and sixfold latency, the dual-view configuration is the main accuracy–cost operating point. This timing conclusion is implementation-specific: batching or parallelizing independent pairs could reduce wall-clock latency without changing the pair count. Likewise,

tolerance to small time and clock perturbations does not establish robustness to unknown long delays, dropped frames, or independently varying frame rates.

The front-end experiments delimit the generalization claim. The architecture is estimator-agnostic and, after 20 epochs of adaptation on training and validation data, improves over canonical averaging on every tested front end, whether adapted to one target or to a mixture; a single mixed model serves all four. What does not transfer is a frozen checkpoint: the fusion weights are coupled to the error distribution of the estimator they were trained on, so deploying CanonFuse3D behind a new estimator should include a short adaptation step, and the adaptation trades native-front-end retention against target recovery. The alternative-front-end experiments use ground-truth Unity 2D tracks to isolate 3D lifting; comparisons that include image-detection noise for those estimators remain necessary.

The image-level study broadens the native-SAM3D evidence to include synthetic RGB occlusion and upstream detection failures. Gains in 15 of 18 conditions support a useful operating range, while the three random ratio-1.0 failures show that an unoccluded second view alone is insufficient to ensure an improvement over averaging. The masking study uses Unity projections to place artificial patches and one frozen SAM3D front end. It does not establish robustness to natural occluders, weather, motion blur, or image-level errors from alternative estimators. Those settings require separate evaluation, and the observed association between detection failures and fusion degradation does not identify a causal mechanism.

Motion blur can affect the refinement through a different error process. During an exposure, fast limb motion or camera motion spreads image evidence along a trajectory, which can reduce joint-localization reliability or cause the front end to miss the person. The resulting 3D errors may persist across adjacent frames, altering both velocity features and cross-view disagreement. A clearer second view may provide useful evidence, but similar blur-induced errors in both streams can leave the gate with little reliable information to select. Errors in the hip or neck landmarks used for the batch anchor can additionally perturb the coordinate transform. These are expected mechanisms, not measured blur effects in this study. Temporal refinement may suppress isolated jitter, but neither temporal smoothing nor the reported patch-occlusion results establish robustness to exposure-induced motion blur.

The reference regimes constrain interpretation. Unity provides an avatar

skeleton in dataset coordinates that is transformed using the same batch-referenced rule; it is not a metric motion-capture measurement. Ski-PTZ-Pose is evaluated against a transformed fused reference derived from view-specific annotations; its result measures agreement with that constructed reference rather than independent physical 3D accuracy. Raw-space and transformed-space errors are therefore not interchangeable, no skeleton-scale normalization is performed, and none of these values should be relabeled as millimeters.

The body reference is shared within each evaluated batch, so changing its composition can change the representation. The results do not establish invariance to regrouping or single-sequence deployment. A sample-independent variant would require retraining and evaluation; the exact reference convention is documented in Supplementary Section S9.

The reliability gate is informative but not calibrated uncertainty. Its mean is explicitly shaped by balance and entropy regularization, so high gate-error correlation and preference accuracy do not establish probabilistic calibration. Because the frozen-checkpoint stress tests reuse one trained checkpoint, they do not quantify variation across independent training runs. Between-dataset uncertainty also remains unresolved. External sports datasets, metric 3D references, longer asynchronous sequences, and naturally occurring image degradation remain necessary directions.

7. Conclusion

CanonFuse3D shows that human-centric fusion can refine two independently estimated 3D pose streams without camera parameters. Pose alignment, cross-view temporal features, and gated residual correction reduce the position error relative to canonical averaging by 44.5% on Unity and 48.6% on Ski-PTZ-Pose and outperform every non-learned fusion baseline in position, Procrustes-aligned, and acceleration error, while a temporal convolutional baseline is competitive in position error and a temporal smoother trades position accuracy for temporal smoothness. The benefit persists across camera separations, small timing and sampling-rate errors, and structured pose or image occlusion, and it is reversed only by dense corruption of both inputs. The same architecture adapts to alternative upstream estimators with lightweight fine-tuning and extends to three and four cameras through pairwise composition, with two views remaining the most favorable accuracy–cost operating point. The findings support calibration-free post-estimation dual-

view fusion as a practical component for distant capture of rapidly moving subjects, while emphasizing that a frozen fusion model should be validated, and if necessary adapted, for the chosen pose estimator and capture conditions.

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Declaration of competing interest

Author confirmation required before submission. The authors must approve the final declaration wording.

CRediT authorship contribution statement

Author confirmation required before submission. Contributions must be approved by all seven authors.

Data and code availability

Ski-PTZ-Pose is available from the dataset provider [21]. The release status and repository URL for the Unity data, trained checkpoint, evaluation manifests, and evaluation scripts must be confirmed by the authors before submission. The author-collected indoor and outdoor videos are used only for qualitative illustration; their sharing conditions must also be confirmed before submission. The local reproducibility package records exact checkpoint and estimator hashes.

Ethics and generative-AI disclosure

The quantitative study uses a synthetic dataset and a previously released research dataset. The author-collected indoor and outdoor videos are used solely for qualitative illustration and do not contribute training samples or quantitative measurements. The authors must confirm and report any applicable institutional-review and participant-consent requirements for these recordings before submission. Journal-required disclosure of generative-AI assistance must also be reviewed and approved by the authors before submission.

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